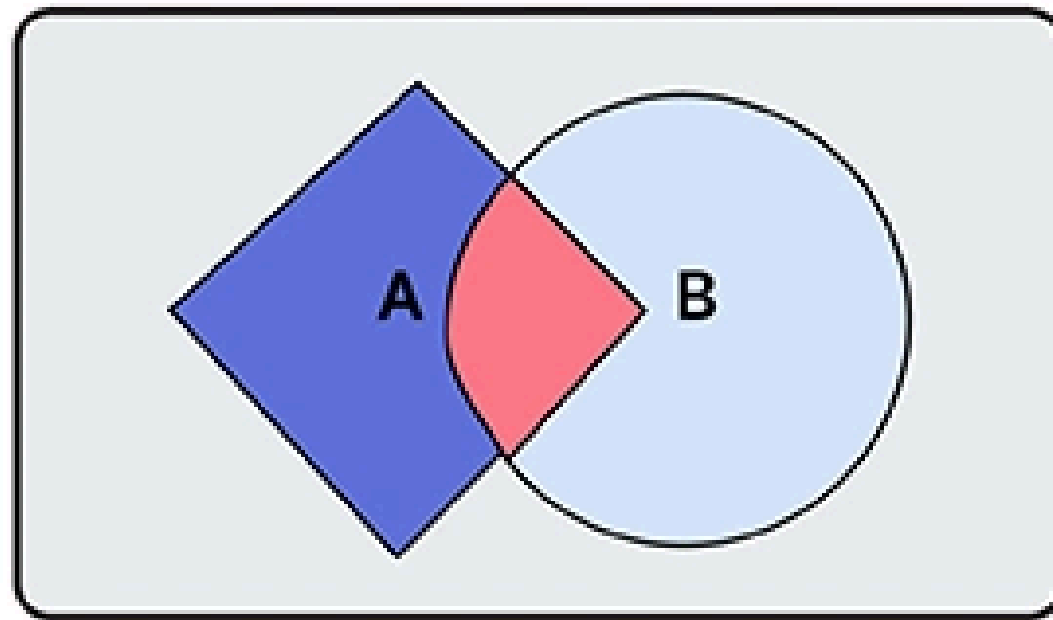


Definition: Bayes Theorem - $P(A|B)$

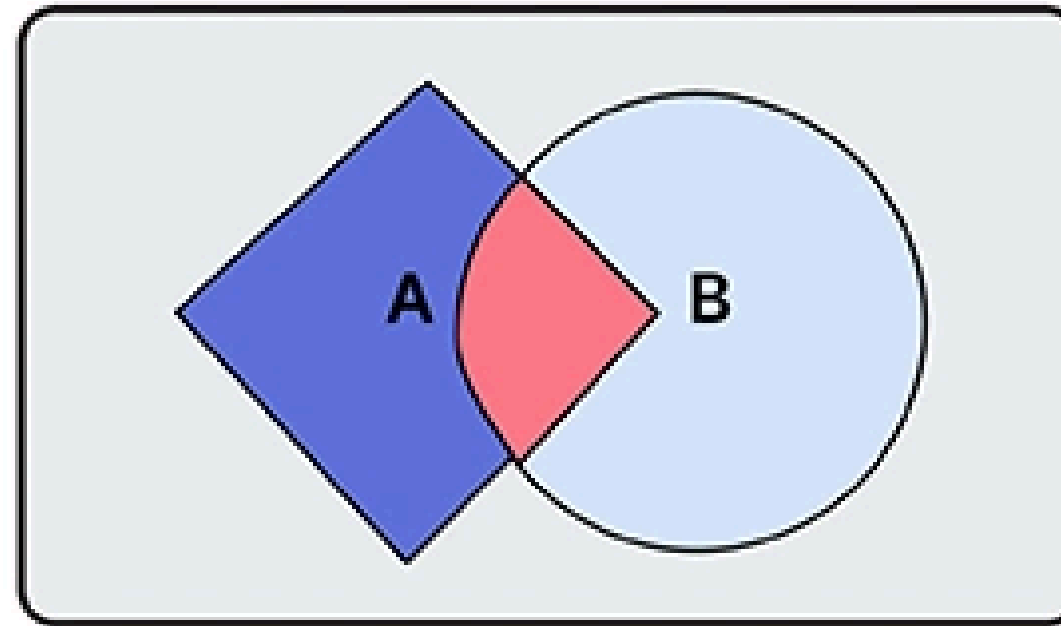


$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

← Using the Law of Total Probability

Visual proof of Bayes' Theorem!

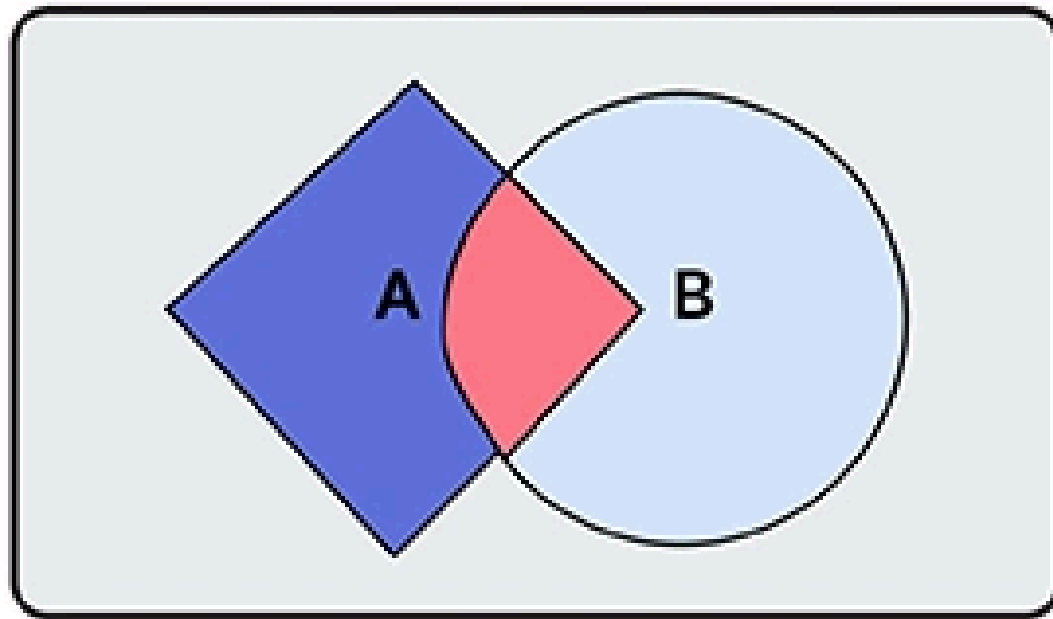


$$P(A) = \frac{\text{blue diamond}}{\text{gray rectangle}} \quad P(B) = \frac{\text{light blue circle}}{\text{gray rectangle}} \quad P(A|B) = \frac{\text{red triangle}}{\text{light blue circle}} \quad P(B|A) = \frac{\text{red triangle}}{\text{blue diamond}}$$

$$\frac{\text{red triangle}}{\text{light blue circle}} = \frac{P(A|B)}{P(B)} = \frac{P(B|A) * P(A)}{P(B)} = \frac{\frac{\text{red triangle}}{\text{blue diamond}} * \frac{\text{blue diamond}}{\text{gray rectangle}}}{\frac{\text{light blue circle}}{\text{gray rectangle}}} = \frac{\text{red triangle}}{\text{light blue circle}}$$

Bayes Theorem

Definition: Bayes Theorem - $P(A|B)$



Bayes Theorem

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

Why is Bayes Theorem Important?

- Updating Beliefs based on new evidence
- Correct for Testing Errors
- Basis for Bayesian Statistics
- Used in a wide array of fields Medical Diagnosis, Medical Trials, Spam Filtering, Finance, etc.

Example: Bayes Theorem

1. Medical Diagnosis

Scenario:

A doctor wants to determine the probability that a patient has a rare disease, given a positive test result. The test is not perfect and can give false positives.

Example:

Suppose:

- **Prevalence (Prior Probability):** 1 in 10,000 people have the disease (0.01%).
- **Test Accuracy:**
 - **Sensitivity (True Positive Rate):** 95% (if you have the disease, the test is positive 95% of the time).
 - **Specificity (True Negative Rate):** 99% (if you do not have the disease, the test is negative 99% of the time).

Calculation:

Let's find the probability a patient has the disease given a positive test result ($P(\text{Disease}|\text{Positive})$):

$$P(\text{Disease}|\text{Positive}) = \frac{P(\text{Positive}|\text{Disease}) \times P(\text{Disease})}{P(\text{Positive})}$$

Where:

- $P(\text{Positive}|\text{Disease}) = 0.95$
- $P(\text{Disease}) = 0.0001$
- $P(\text{Positive}) = P(\text{Positive}|\text{Disease}) \times P(\text{Disease}) + P(\text{Positive}|\text{No Disease}) \times P(\text{No Disease})$
 - $P(\text{Positive}|\text{No Disease}) = 1 - 0.99 = 0.01$
 - $P(\text{No Disease}) = 0.9999$

So:

$$P(\text{Positive}) = (0.95 \times 0.0001) + (0.01 \times 0.9999) \approx 0.000095 + 0.009999 = 0.010094$$

Now plug into Bayes' Theorem:

$$P(\text{Disease}|\text{Positive}) = \frac{0.95 \times 0.0001}{0.010094} \approx \frac{0.000095}{0.010094} \approx 0.0094$$

Example: Bayes Theorem

2. Spam Filtering

Scenario:

An email service wants to classify emails as spam or not spam based on the presence of certain keywords.

Example:

Suppose:

- **Prior Probability:** 20% of all emails are spam ($P(\text{Spam}) = 0.2$).
- **Keyword:** “Free” appears in 50% of spam emails ($P(\text{“Free”}|\text{Spam}) = 0.5$) and in 5% of non-spam emails ($P(\text{“Free”}|\text{Not Spam}) = 0.05$).

Calculation:

Find the probability an email is spam given it contains the word “Free” ($P(\text{Spam}|\text{“Free”})$):

$$P(\text{Spam}|\text{“Free”}) = \frac{P(\text{“Free”}|\text{Spam}) \times P(\text{Spam})}{P(\text{“Free”})}$$

Where:

- $P(\text{“Free”}) = P(\text{“Free”}|\text{Spam}) \times P(\text{Spam}) + P(\text{“Free”}|\text{Not Spam}) \times P(\text{Not Spam})$
 - $P(\text{Not Spam}) = 0.8$
 - $P(\text{“Free”}) = (0.5 \times 0.2) + (0.05 \times 0.8) = 0.1 + 0.04 = 0.14$

Now plug into Bayes’ Theorem:

$$P(\text{Spam}|\text{“Free”}) = \frac{0.5 \times 0.2}{0.14} = \frac{0.1}{0.14} \approx 0.714$$